

SUSHANT PAUDYAL

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Education

Islington College

BSc (Hons) Computing

Sep 2022 – Oct 2025

Kamal Pokhari, Kathmandu

Experience

Sangalo Tech P. Ltd

Front End Developer Intern

Sep 2024 – Nov 2024

Lokanthali, Bhaktapur

- Implemented frontend components using **React.js** and modern web technologies to ensure clean, user-friendly experiences.
- Collaborated with designers and developers to translate design concepts into functional interfaces.
- Gained hands-on experience working in a professional development environment, improving both technical and teamwork skills.

Projects

KhelaSathi-system | *React.js, Node.js, Express.js, PostgreSQL, Docker*

- Developed a full-stack futsal booking website using the **PERN stack** with a responsive UI styled using Tailwind CSS.
- Implemented **real-time slot booking and cancellation** through a custom-built REST API.
- Integrated Khalti payment gateway** for secure digital transactions and email notifications for booking updates.
- Designed a **multi-tenancy architecture** with role-based portals for admins, venue owners, and players.
- Containerized the application using **Docker Compose** for consistent development and testing environments.

Optimizing Container Image Efficiency | *Docker*

- Explored and implemented **multi-stage Docker builds** to improve container image efficiency.
- Reduced image size by **30–65%**, resulting in faster build and startup times.
- Optimized Dockerfiles by removing unnecessary dependencies and files.
- Gained hands-on experience with containerization and image management using Docker.

Automated Database Backup System | *Linux, Bash*

- Designed and configured a two-server architecture consisting of a **Database Server** and a **Backup Server** with restricted IP-based access control.
- Automated daily database backups using **Bash scripting and cron jobs**, including archiving and secure transfer to the backup server.
- Implemented secure backup encryption using **AES for file encryption** and **RSA for encrypting the AES key**.
- Configured **SSH key-based authentication** for secure communication between servers.
- Automated extraction, decryption, and push of backups to a remote **GitHub repository via SSH**, with retention policies (2 days on DB server, 7 days on backup server).

IoT Based Smart System | *Arduino, Sensors, C/C++*

- Developed an IoT prototype using microcontrollers and sensors to collect real-time environmental data.
- Built and integrated **gas leakage sensor** and **obstacle-avoiding sensor** for enhanced safety and automation.
- Implemented sensor communication and control logic for monitoring and real-time responses.
- Gained practical knowledge of **embedded systems, hardware interfacing, and IoT architecture**.

Basic Robotics and Automation Experiments

- Explored robotics fundamentals including sensors, microcontrollers, and automation logic.
- Learned hardware interfacing concepts and basic robotics programming.
- Built small experimental prototypes to understand robotic movement and control systems.

Achievements

Bhabishya STEAM Challenge Participant

- Participated in the **Bhabishya STEAM Challenge**, gaining exposure to robotics, innovation, and problem-solving activities.
- Collaborated with team members to develop creative solutions using **STEM and robotics concepts**.

Technical Skills

Languages: C/C++, Python, Java, JavaScript, SQL, HTML, CSS
Robotics & IoT: Arduino, Sensors, Basic Embedded Systems, IoT Concepts
Frameworks & Libraries: React.js, Express.js
DevOps & Containerization: Docker, Kubernetes (k3s), Jenkins
Tools: Git, Postman, Bash, Vim
Operating System: Linux, macOS, Windows

Training & Certifications

MERN Stack Training	Jun 2024 – Sept 2024
<i>Broadway Infosys</i>	<i>Tinkune, Kathmandu, Nepal</i>
- Completed hands-on training covering MongoDB, Express.js, React.js, and Node.js for full-stack web development. - Built responsive frontend components and integrated backend APIs using modern JavaScript practices.	

DevOps Training	Dec 2025 – mar 2026
<i>Broadway Infosys</i>	<i>Tinkune, Kathmandu, Nepal</i>
- Training in Linux administration, Git/GitHub, CI/CD fundamentals, and deployment workflows. - Gaining practical exposure to server setup, automation, and DevOps best practices.	